

1. Please work through the notes for cross-section calculations which have been uploaded under the "Files" tab.
2. Consider a particle of mass m decaying into three massless particles ($k_1 \rightarrow k_2 + k_3 + k_4$). You are given that in a certain field theory, the matrix element squared for the process is given by:

$$|M|^2 = 64g^2(k_1 \cdot k_3)(k_2 \cdot k_4), \quad (1)$$

where, $k_1^2 = m^2, k_2^2 = k_3^2 = k_4^2 = 0$. The differential decay rate in the rest frame of the decaying particle is then,

$$d\Gamma = (2\pi)^4 \delta^4(k_1 - k_2 - k_3 - k_4) \frac{|M|^2}{2m} \frac{d^3k_2}{(2\pi)^3 2k_2^0} \frac{d^3k_3}{(2\pi)^3 2k_3^0} \frac{d^3k_4}{(2\pi)^3 2k_4^0}. \quad (2)$$

Find the total decay rate.

3. In formula (1) of the notes given in class for cross section calculations, in a given theory,

$$|M|^2 = e^4 \left\{ \left(\frac{p \cdot k}{p \cdot k'} + \frac{p \cdot k'}{p \cdot k} \right) + 2m^2 \left(\frac{1}{p \cdot k} - \frac{1}{p \cdot k'} \right) + m^4 \left(\frac{1}{p \cdot k} - \frac{1}{p \cdot k'} \right)^2 \right\}. \quad (3)$$

Here p and k are the incoming particles and p' and k' are the outgoing ones with, $p^2 = p'^2 = m^2; k^2 = k'^2 = 0$. If ω and ω' denote the energies k^0 and k'^0 in the lab frame, show that the differential cross section in the lab frame may be written as:

$$\frac{d\sigma}{d\Omega} = \frac{e^4}{(32\pi^2 m^2)} \left(\frac{\omega'}{\omega} \right)^2 \left\{ \frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2\theta \right\}, \quad (4)$$

where,

$$\omega' = \frac{m\omega}{m + \omega(1 - \cos\theta)} \quad (5)$$